

1 Small pilot assay

pilot perturbation cellWT / control

25-50 cells per perturbation

2 Estimate measurement properties

variant-to-WT effect size

within-variant split-half noise

half 1 half 2

pilot effect size-to-noise estimate

3 Predict rankability

likely to clear the measurement floor?

likely measurable

likely false

proceed to full-depth benchmarking

increase the depth, or measurement, or both

predicts benchmark reliability, not experiment noise or downstream model performance

Can a small pilot identify perturbations whose ground truth is likely to become measurable at higher depths?

